BUSINESS INTELLIGENCE I | 2026-2027

Inside Airbnb
Initial data analysis

Lisbon-region data | Project readiness, statistical exploration and a proposed sampling strategy

Prepared 20 October 2026. Based on the full raw-data profile already completed; proposed samples and Fabric implementation have not been executed.

25,020 listings	1,950,076 review rows	9,132,343 calendar rows
Current property snapshot	Historical review activity	Prospective availability

Overall conclusion

A promising source, but not yet a compliant project design. The data combines rich listing attributes, long review history and calendar availability. Listings meet the source row cap. Reviews and calendar exceed it, while the required fact grain, two primitive measures and five hierarchies still need a defensible design.

What the analysis established

• All rows in seven logical files and 128 fields were profiled. Original gzip payloads match extracted CSVs; raw-file hashes were unchanged.

• Prices are missing for 2,977 listings (11.90%). Among 22,043 observed prices, the mean is 194.58 and the median is 140.

• Twelve detailed-listing fields are entirely empty. Review-related blanks occur in 2,822 listings with zero reviews.

• 443 review IDs have conflicting row variants, producing 1,147 extra repeated-ID rows. These are not exact duplicate rows.

• No calendar/review listing keys are orphaned. Stay rules conflict in 508 calendar rows and one listing.

How to read this report

The next pages separate observed evidence, source limitations and proposed choices. Charts are descriptive, not causal. The report ends with sampling options, example business questions, numerical reference tables and sources.

01 / DATA FOUNDATION

Three grains, three different meanings

Data	One row represents	What it can establish
Listings	A listing observed at a collection date	Current advertised attributes and source-modelled metrics
Reviews	An observed review event	Historical review activity, not all stays or payments
Calendar	A listing and prospective date	Availability and stay restrictions, not actual bookings

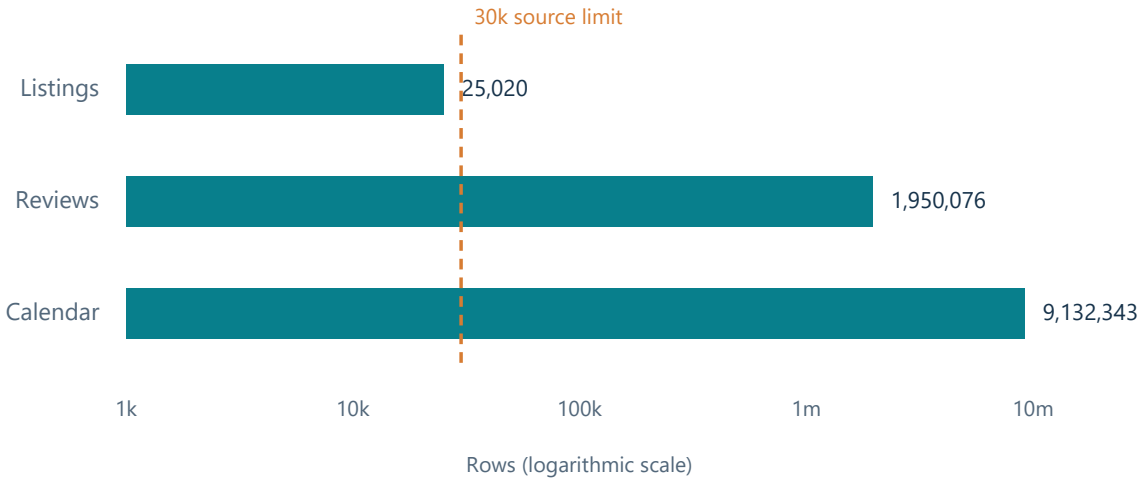


Figure 1. Detailed CSV row counts on a logarithmic scale. The dashed line marks the assignment source cap of 30,000 rows per dataset. Gzip/extracted pairs are the same payload, not extra sources.

Additional file	Rows / features	Interpretation
Summary listings	25,020	Overlaps detailed listings; do not append as extra listings.
Summary reviews	1,950,076	Only listing/date; cannot uniquely identify review events.
Neighbourhood CSV	134	Compound municipality/neighbourhood lookup.
Neighbourhood GeoJSON	173 features / 134 keys	Multiple shapes may share one geographical key.

Collection dates span 19-22 September 2026. The labels cover the wider Lisbon region, not only Lisboa municipality. All 131 geography pairs used by listings are covered by the neighbourhood lookup and GeoJSON.

Interpretation limit: coordinates are approximate; an unavailable calendar night can be booked or host-blocked. Current listing attributes should not be treated as historical attributes of older review events. [S2-S4]

01A / WORKED RELATIONSHIP EXAMPLE

One property, several calendar dates and reviews

The following records were checked directly in the raw files. They all refer to listing **6499**. Calendar.listing_id and Reviews.listing_id connect to Listings.id. The property information is a current snapshot; review dates and calendar dates describe different things.

1. The parent listing

id	host_id	Room type	Municipality	Neighbourhood	Collected
6499	14455	Entire home/apt	Lisboa	Belm	2026-09-20

Source: listings.extracted.csv. "Belm" is the raw neighbourhood label, reproduced without silently correcting it. This is an example of a label to investigate during cleansing.

2. Calendar rows linked to that listing

listing_id	date	available	minimum_nights	maximum_nights
6499	2026-09-20	t (true)	3	1125
6499	2026-09-21	t (true)	3	1125

Each row describes a prospective listing-date. Joining these rows to Listings on listing_id = id adds the room type, municipality and host without creating additional calendar rows, provided the listing key is unique.

3. Historical reviews linked to the same listing

Review id	listing_id	Review publication date
18879225	6499	2014-09-02
21074122	6499	2014-10-11

These reviews connect to the same property. They do **not** identify stays on the September 2026 calendar dates. The source provides no booking identifier or stay-date mapping to establish that relationship.

These are selected example rows, not a claim that the listing has only two calendar records or only two reviews. Any use of the 2026 room type to group 2014 reviews must be labelled as grouping by the listing's current attributes, not its verified 2014 state.

01B / AVOIDING DOUBLE COUNTING

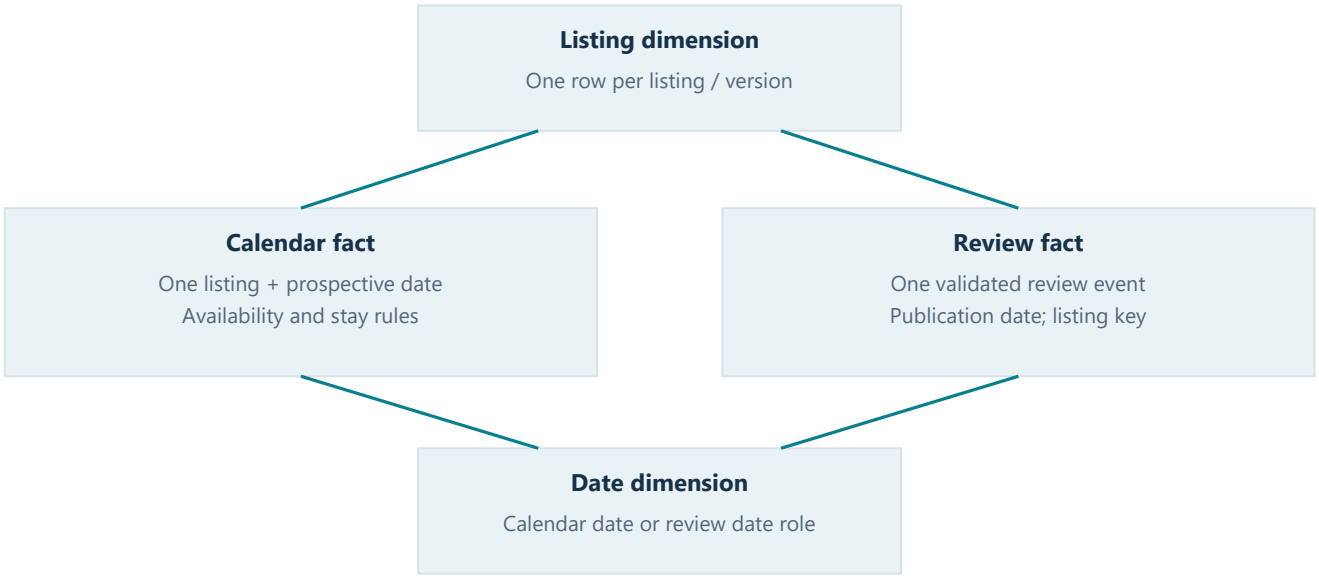
Why a direct calendar-to-review join is misleading

If the two calendar rows above are joined to the two review rows using only listing_id, every calendar row matches every review row for that listing. The result has **2 x 2 = 4 rows**:

listing_id	Calendar date	Review id	Review date
6499	2026-09-20	18879225	2014-09-02
6499	2026-09-20	21074122	2014-10-11
6499	2026-09-21	18879225	2014-09-02
6499	2026-09-21	21074122	2014-10-11

Each review is repeated twice and each calendar day is repeated twice. For an illustrative listing with 365 calendar rows and 100 reviews, the same join would produce **36,500 rows**. Counting or summing on that result inflates the original events and availability observations.

Proposed warehouse pattern: keep two fact grains



Each dimension links one-to-many to each fact; the facts do not join directly to one another at row level. The listing/day and review-event grains remain distinct. This diagram is a design proposal, not an implemented or approved Fabric schema.

For a combined analysis, aggregate each fact separately to the same justified comparison level, then join the aggregates. For example, compare annual review counts with explicitly defined listing-level snapshot measures. A shared Date dimension does not make a review publication date equal to a stay date.

Data checks found zero orphan listing IDs in calendar and reviews and zero duplicate detailed-listing IDs. Review-event identity still needs resolution for the 443 IDs with conflicting variants. Separate fact tables do not by themselves settle the assignment's primitive-measure or hierarchy requirements.

02 / MARKET COMPOSITION

Supply is concentrated in entire homes and Lisboa

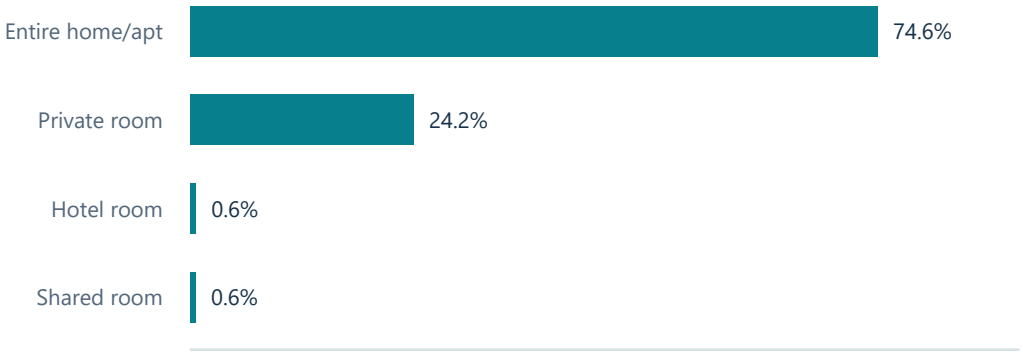


Figure 2. Listing composition by room type; denominator 25,020 listings. Entire home/apt: 18,658; private room: 6,065; hotel room: 152; shared room: 145.

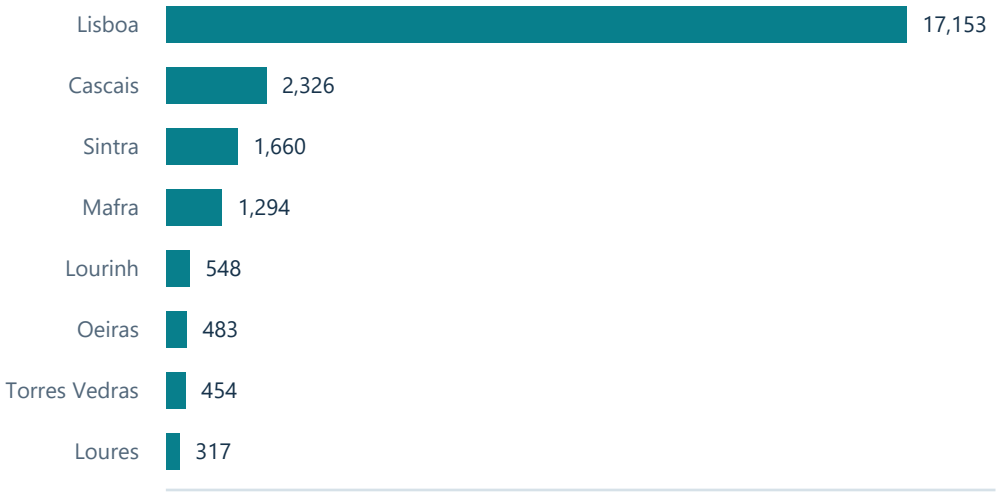


Figure 3. Eight largest municipality groups by listing count. Source labels are retained; some labels contain spelling/diacritic inconsistencies.

Lisboa accounts for 68.6% of listings. An unstratified sample may underrepresent smaller municipalities and the two rare room categories. Preserve these groups explicitly when sampling if geographic or accommodation comparisons matter.

Geography has 131 distinct municipality/neighbourhood combinations in listings. One neighbourhood label occurs in multiple groups: use both group and neighbourhood as the business key before assigning a warehouse surrogate key.

These are counts of listings in this snapshot, not counts of dwellings across the entire housing market or historical market shares.

03 / ADVERTISED PRICES

The mean is pulled upward by a long price tail

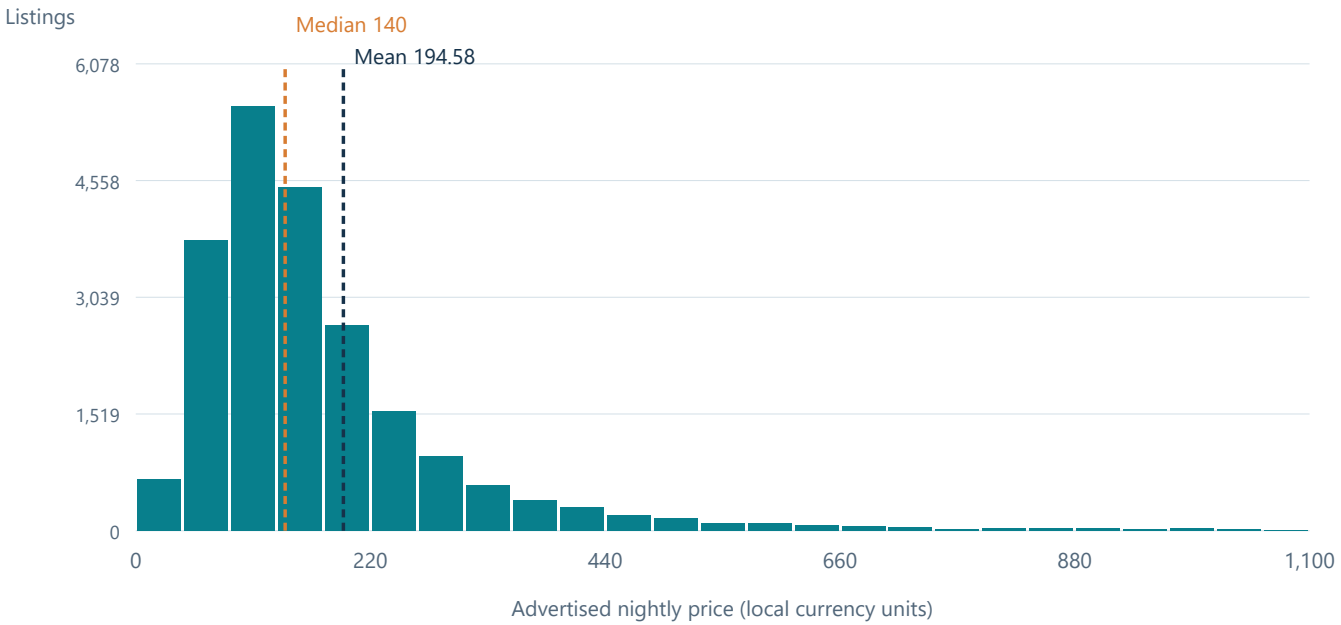


Figure 4. Price histogram for observed prices up to 1,100; 199 higher values are outside this visual window. All 22,043 nonmissing prices, including the tail, are included in the mean, median and outlier analysis.

Valid prices	Missing prices	Mean	Median	Range
22,043	2,977 (11.90%)	194.58	140.00	14.00-13,610.67

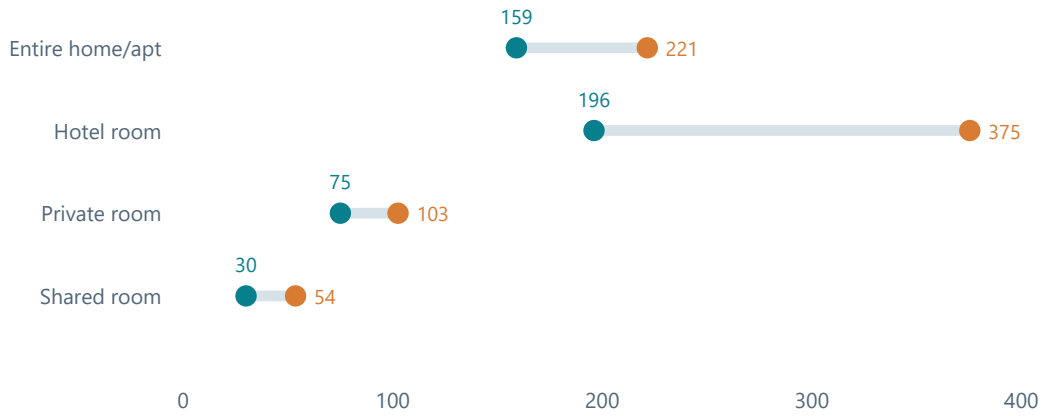


Figure 5. Median (teal) versus mean (orange) advertised prices by room type. Means exceed medians in every category. Hotel/shared-room groups are much smaller than entire-home/private-room groups.

The median is a useful baseline for a typical advertised listing because it is less sensitive to extreme prices. The higher mean signals skew, not proof of erroneous records. Comparisons should account for room type, capacity and location.

Price units are the source export's local currency units. A currency symbol alone does not establish the currency. These are advertised amounts, not realized booking prices or revenue. [S3]

04 / COMPLETENESS

Missing values have different explanations

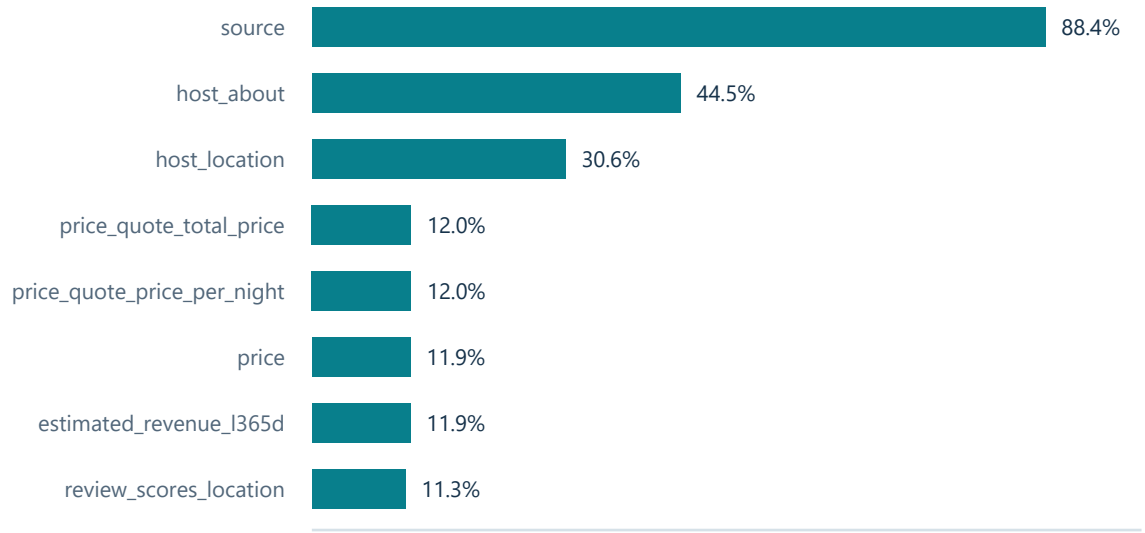


Figure 6. Eight highest missing shares among partially populated detailed-listing fields. The 12 entirely empty fields are listed separately below. Empty CSV cells are missing; whitespace-only cells were counted separately.

Entirely empty fields (12)	
neighborhood_overview	host_neighbourhood
host_since	host_total_listings_count
host_response_time	neighbourhood
host_response_rate	bathrooms
host_acceptance_rate	calendar_updated
host_thumbnail_url	instant_bookable

Observed patterns, not blanket imputation

All 2,822 missing values in each of rating, first review, last review and reviews per month occur in listings with zero reviews. This supports handling review-related absence separately from price or host-information gaps. A missing rating should not be replaced by a zero rating.

Preserve the raw values. In staging, define field-specific policies: exclude entirely empty fields from the working model unless needed, retain missing-price indicators, and distinguish unknown values from true zero counts. Do not invent host registration dates or infer data that the export does not provide.

05 / EXTREMES AND DATA RULES

Statistical outliers are candidates for review

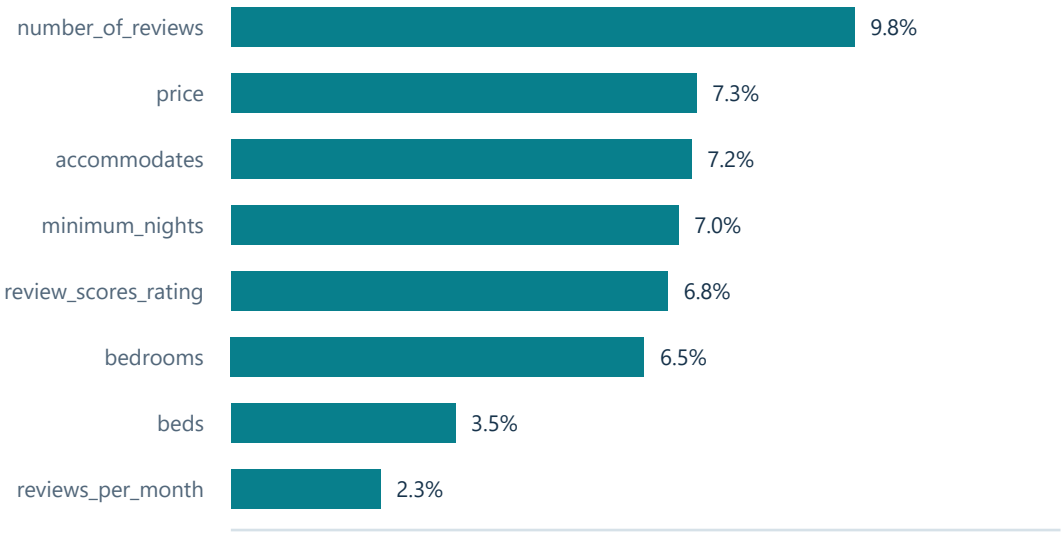


Figure 7. Potential outliers as a share of valid numeric observations, using values outside Q1 - 1.5 x IQR or Q3 + 1.5 x IQR. Different fields have different distributions; rates are not a ranking of data quality.

Feature	Minimum	Maximum	IQR flags
price	14.00	13,610.67	1,608
bedrooms	0.00	43.00	1,617
beds	1.00	45.00	866
minimum_nights	1.00	730.00	1,751
maximum_nights	1.00	10,000.00	1

Separate statistical extremes from rule conflicts

A large property, a high-volume host or a long maximum stay may be legitimate. Even a low aggregate rating can be statistically unusual without being invalid. Review source context before capping, removing or transforming values.

Rule conflicts are more direct: minimum stay exceeds maximum stay in 508 calendar rows and one listing. Flag these records in staging, inspect the original fields and document the chosen treatment. The local rule checks are proposals, not confirmation of the four checks required in the labs.

06 / TIME AND INTEGRITY

Review history is long, but it is not booking history

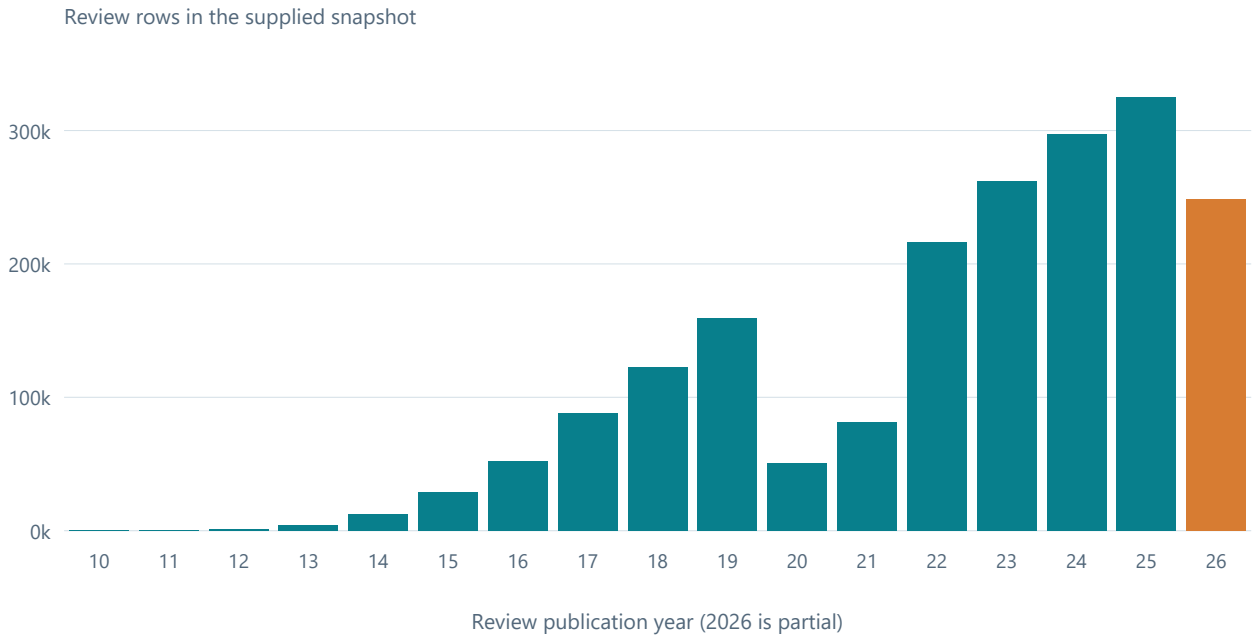


Figure 8. Full-data review row counts by year. 2026 is partial through 21 September, and 2010 begins on 24 July. Counts retain conflicting review-ID variants pending resolution; they describe the supplied snapshot.

Reviews span 24 July 2010 to 21 September 2026, satisfying the date-span aspect of the more-than-one-year requirement. However, reviews are only observed feedback events. Their volume does not establish total stays, guests, occupied nights or revenue.

The calendar covers 19 September 2026 to 21 September 2027 across collection dates. Its 367-day overall span is prospective and reflects staggered collection; it does not supply multi-year historical bookings. Current-snapshot selection and deleted listings can also affect historical coverage. [S4]

Integrity check	Observed result	Implication
Review identifiers	443 IDs; 1,147 extra repeated-ID rows; zero exact duplicate rows	Investigate conflicting variants; do not blindly drop repeated IDs.
Listing references	Zero orphan listing IDs in calendar and review files	Joins to the detailed listing table are feasible.
Calendar grain	Zero repeated listing/date keys	A listing-day grain is supported by the key check.
Geography	Zero missing lookup pairs	Retain compound keys; do not count shapes as extra places.

07 / ASSIGNMENT READINESS

What is established, and what still needs work

Requirement	Assessment	Next evidence needed
Source size <=30k	Listings pass; reviews/calendar exceed the cap	Document linked samples without changing originals.
Source >=10k fact rows; DW >=5k	Volumes exist, but valid final fact rows are unconfirmed	Declare grain; resolve ID conflicts; check counts after selection.
History >1 year	Review dates support it; calendar is prospective	Confirm review events are accepted for the chosen scenario.
At least 5 dimensions incl. Date	Date, Listing, Host, Geography, Accommodation Type are candidates	Define keys, attributes and business roles.
At least 5 hierarchies; average depth 3	Not demonstrated by selecting five dimensions	Design meaningful roll-ups; obtain teacher clarification if needed.
At least 2 primitive measures	Not established at a historical review-event grain	Find suitable observed measures at one grain, or supplement sources.
Fabric staging, DW and ETL	Not verified by local profiling	Separate staging database and STG/DW pipelines; Copy Data and Dataflow.
DQ logging and report	Exploration supplies evidence, not completed delivery	Implement four lab-defined DQ checks and staging LOG; complete report.

The key modelling issue

A review row offers an event count of one, but it has no payment amount or individual rating here. Current listing price, capacity and aggregate ratings belong to a different grain and time. Attaching them to older reviews does not make them historical transaction measures. Text length and identifiers do not solve the primitive-measure requirement.

A listing-day availability fact is a different possible scope, with observed availability and stay-rule values. Whether these qualify for the intended transactional project and primitive-measure expectations needs teacher agreement. Source-estimated occupancy/revenue are model outputs, not observed transactions.

Assignment references: guide v1.1, pp.2-6. Source and warehouse row minimums appear in different sections; confirm their intended application. Dashboarding is beyond the initial BI I warehouse/ETL scope. [S1]

08 / PROPOSED SAMPLING

Reduce rows while preserving interpretable links

Dataset	Illustrative proposal	Conditions
Detailed listings	Keep all 25,020	Already below 30k; supplies parent keys.
Detailed reviews	25,000 valid events	Resolve conflicting IDs first; stratify by year, municipality and room type.
Calendar	Up to 80 listings x 365 dates = 29,200	Enforce a 365-date maximum per listing; choose listings deliberately.
Neighbourhoods	Keep the small lookup	Retain referenced keys; separate shapes from attributes.
Summary exports	Reference/reconciliation only	Avoid overlapping records in the analytical fact sources.

Sampling sequence

• Decide business questions, fact grain and acceptance of the proposed measures before fixing sample sizes.

• Investigate review-ID conflicts and define valid events. Allocate a reproducible sample across relevant strata; preserve rare groups explicitly when needed.

• Sample child events first and retain all required listing, host and geography lookups. Do not independently random-sample parent tables.

• Use a fixed seed and save selected event IDs, counts, exclusions and date/category coverage in a sampling manifest.

• Check every submitted table is below 30,000 rows, historical coverage remains adequate, and fact minimums are met after validation.

Keep originals in data/raw and place derived extracts in data/processed. The 80-listing calendar example represents a smaller population than the broad review sample; do not compare the two as if their scope were identical.

This is a proposal, not a sample already generated. If all 131 geography groups must be represented in the calendar sample, 80 listings cannot achieve that; redesign the listing/date allocation. Do not infer actual occupancy from unavailable dates.

09 / ANALYTICAL CHOICES

Column priorities should follow the business questions

Priority	Keep / consider	Example questions
1 - Core structure and activity	Review id/listing_id/date; listing id/host_id, room/property type, cleansed geography, scrape date; calendar listing/date/available	How does review activity vary over time and geography? Where is current supply concentrated? Which future listing-days are available?
2 - Price, capacity and quality	Price, capacity, beds/bedrooms, stay rules, aggregate ratings, review counts/frequency and availability windows	Which groups have higher median advertised prices? How do prices vary with capacity? How do stay rules or aggregate ratings differ by location/type?
3 - Specialised enrichment	Amenities, licence information, superhost flag if populated, text comments and descriptions; verified source estimates	Which amenities are associated with prices? What topics occur in comments? Which records provide registration information?

A possible project scenario

A Lisbon-region accommodation market observatory wants to understand current listing supply, advertised pricing, historical review activity and prospective availability across locations and accommodation categories. This is an example for discussion, not an approved organizational brief.

Respect the evidence behind each question

• Compare prices as advertised values; do not label them revenue. Associations between amenities or host status and prices do not prove causation.

• Use current attributes to describe the current listing snapshot. Historical reviews grouped by current room type must be labelled that way.

• A licence string does not establish legal compliance. Superhost-related questions cannot be assumed feasible until field completeness is checked.

• Year/quarter/month/day and country/municipality/neighbourhood are plausible hierarchy examples, but five dimensions do not automatically provide five hierarchies.

The three priority levels are cumulative column-selection priorities, not separate independently sampled datasets. No columns are removed from raw files.

APPENDIX A / 1

Detailed-listing numerical reference

Full-data statistics; 25,020 listing rows. Missing cells are excluded from numeric calculations. Identifiers, text, dates and booleans have no arithmetic mean or median. Percent fields are percentage points (0-100).

Feature	Missing %	Distinct	Mean	Median	Min	Max
accommodates	0.00	16	3.81	4.00	1.00	16.00
availability_30	0.00	31	9.90	8.00	0.00	30.00
availability_365	0.00	366	227.33	275.00	0.00	365.00
availability_60	0.00	61	27.87	29.00	0.00	60.00
availability_90	0.00	91	49.16	55.00	0.00	90.00
availability_eoy	0.00	105	57.12	65.00	0.00	104.00
bathrooms	100.00	0	N/A	N/A	N/A	N/A
bedrooms	0.19	20	1.67	1.00	0.00	43.00
beds	1.53	30	2.44	2.00	1.00	45.00
calculated_host_listings_count	0.00	68	24.76	5.00	1.00	359.00
calculated_host_listings_count_entire_homes	0.00	55	18.17	2.00	0.00	350.00
calculated_host_listings_count_private_rooms	0.00	38	6.43	0.00	0.00	213.00
calculated_host_listings_count_shared_rooms	0.00	8	0.04	0.00	0.00	7.00
estimated_occupancy_l365d	0.00	93	76.04	42.00	0.00	255.00
estimated_revenue_l365d	11.90	9,052	13,950.41	6,840.00	0.00	1,290,300.00
host_acceptance_rate	100.00	0	N/A	N/A	N/A	N/A
host_listings_count	0.00	122	40.20	6.00	1.00	6,538.00
host_response_rate	100.00	0	N/A	N/A	N/A	N/A
host_total_listings_count	100.00	0	N/A	N/A	N/A	N/A
hosts_time_as_host_months	0.00	12	4.99	5.00	0.00	11.00
hosts_time_as_host_years	0.00	16	6.74	7.00	0.00	15.00
hosts_time_as_user_months	0.00	12	5.08	5.00	0.00	11.00
hosts_time_as_user_years	0.00	18	7.64	8.00	0.00	17.00
latitude	0.00	13,820	38.76	38.72	38.68	39.30

Distinct counts refer to nonempty raw value strings; formatting variants can be distinct before numeric conversion. Means and medians use finite parsed numbers. New quote/tenure/estimate fields have unconfirmed detailed semantics; do not use their names as proof of a measure definition.

APPENDIX A / 2

Detailed-listing numerical reference

Full-data statistics; 25,020 listing rows. Missing cells are excluded from numeric calculations. Identifiers, text, dates and booleans have no arithmetic mean or median. Percent fields are percentage points (0-100).

Feature	Missing %	Distinct	Mean	Median	Min	Max
longitude	0.00	15,004	-9.21	-9.15	-9.49	-8.86
maximum_maximum_nights	0.22	166	592.42	365.00	1.00	10,000.00
maximum_minimum_nights	0.22	70	6.35	3.00	1.00	1,000.00
maximum_nights	0.22	166	592.42	365.00	1.00	10,000.00
maximum_nights_avg_ntm	0.00	1,561	551.57	365.00	1.00	2,125.00
minimum_maximum_nights	0.22	163	524.85	365.00	1.00	2,125.00
minimum_minimum_nights	0.22	56	3.87	2.00	1.00	730.00
minimum_nights	0.22	56	3.87	2.00	1.00	730.00
minimum_nights_avg_ntm	0.00	318	5.01	2.10	1.00	730.00
number_of_reviews	0.00	707	77.94	25.00	0.00	1,933.00
number_of_reviews_l30d	0.00	22	1.14	0.00	0.00	71.00
number_of_reviews_ltm	0.00	143	13.30	6.00	0.00	418.00
number_of_reviews_ly	0.00	141	12.98	4.00	0.00	357.00
price	11.90	5,953	194.58	140.00	14.00	13,610.67
price_quote_price_per_night	11.96	5,943	194.59	140.00	14.00	13,610.67
price_quote_total_price	11.96	5,500	629.23	289.00	14.00	240,652.00
review_scores_accuracy	11.28	158	4.72	4.82	0.00	5.00
review_scores_checkin	11.28	153	4.78	4.88	0.00	5.00
review_scores_cleanliness	11.28	170	4.68	4.80	0.00	5.00
review_scores_communication	11.28	144	4.79	4.89	1.00	5.00
review_scores_location	11.28	165	4.71	4.80	1.00	5.00
review_scores_rating	11.28	162	4.66	4.77	1.00	5.00
review_scores_value	11.28	170	4.57	4.67	1.00	5.00
reviews_per_month	11.28	814	1.58	1.09	0.01	41.57

Distinct counts refer to nonempty raw value strings; formatting variants can be distinct before numeric conversion. Means and medians use finite parsed numbers. New quote/tenure/estimate fields have unconfirmed detailed semantics; do not use their names as proof of a measure definition.

Calendar rules: all 9,132,343 listing-days

Feature	Mean	Median	Min	Max
minimum_nights	5.01	2.00	1.00	1,000.00
maximum_nights	551.09	365.00	1.00	10,000.00

APPENDIX B / METHODS AND SOURCES

What was measured, and how to reproduce it

Coverage and statistical definitions

The local analysis processed all rows in six CSVs and all GeoJSON features in chunks of 100,000 rows. Temporary SQLite frequency tables supported exact distinct counts. Gzip files were verified against extracted payloads and not profiled as additional logical datasets. Raw-file SHA-256 hashes matched before and after analysis.

Empty CSV cells count as missing; whitespace-only cells are tracked separately. NA-like strings were not automatically replaced. Currency/percentage formatting and thousands separators were removed for numeric parsing. IDs remained strings to preserve large integer identifiers. Non-numeric mean/median values are not applicable.

Quartiles and medians use exact linear interpolation over full-data value frequencies. IQR is Q3 minus Q1. Potential outliers are values below Q1 - 1.5 x IQR or above Q3 + 1.5 x IQR. These flags are not a deletion policy. Listing counts, duplicate IDs and price statistics were cross-checked against independent pandas calculations.

Calendar numerical values are stay-length restrictions in nights, not observed lengths of stay. No calendar fields were missing. Summary listing numerical fields overlap the detailed listing export and are not duplicated in the appendix.

Reproducibility in the BI Project folder

The saved profiler is scripts/profile_raw_data.py, with tested pandas 3.0.1 and numpy 2.3.5. Its outputs include the full 128-field profile, category counts, file hashes, key checks and date ranges. The local exploration is not the submitted Microsoft Fabric ETL, and it does not establish that the four lab checks have been implemented.

Sources and evidence

[S1] Business Intelligence I Project Guide 2026-2027 v1.1, dated 13 October 2026. Local PDF in docs/source/. Requirements: pp.2-6; report: pp.7-8; delivery: p.9.

[S2] Inside Airbnb, Get the Data. File roles and licence. Source data attribution: Inside Airbnb, CC BY 4.0 as listed in the catalogue.

[S3] Inside Airbnb, Data Dictionary, public legacy version 4.3 (2022). It does not explain every newer field in these exports; unverified interpretations remain hypotheses.

[S4] Inside Airbnb, Data Assumptions. Snapshot, availability, approximate-location and model-estimate caveats.

[S5] Saved full-data evidence: outputs/raw-data-analysis/profile.json, column_profile.csv, room_type_summary.csv, geography_summary.csv and reviews_by_year.csv. Analysis and source consultation: 20 October 2026.

Limitations: the exact download date and historical download URLs of local files were not supplied. Statistical results describe the supplied exports, including unresolved review-ID conflicts. No teacher approval, business scenario selection, sampling execution, Fabric load or final delivery is asserted.